

Solved Problems (L'Hopital)

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Chapter II: Indeterminate Forms, L' Hospital's rule

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Evaluate the following:

$$(i) \lim_{x \rightarrow 0} \frac{x + \sin 2x}{x - \sin 2x} \quad (ii) \lim_{x \rightarrow 0} \left(\frac{4}{x^2} - \frac{2}{1 - \cos x} \right) \quad (iii) \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x}}$$

$$(iv) \lim_{x \rightarrow 0} \frac{x - \sin^{-1} x}{\sin^3 x} \quad (v) \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x} \quad (vi) \lim_{x \rightarrow 0} \frac{x - \sin x}{\tan^3 x}$$

$$(vii) \lim_{x \rightarrow 0} (a^x + x)^{\frac{1}{x}} \quad [\text{Hints: } \frac{d}{dx}(a^x) = a^x \log_e a, \text{ where } a \text{ is a positive constant}]$$

$$(viii) \lim_{x \rightarrow 0} \frac{\ln \ln(1 - x^2)}{\ln \ln \cos x} \quad (ix) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sec^2 x - 2 \tan x}{1 + \cos 4x} \quad (x) \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$$

$$(xi) \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}} \quad (xii) \lim_{x \rightarrow 0} \frac{(e^x - 1) \tan^2 x}{x^3} \quad (xiii) \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}}$$

$$(xiv) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right) \quad (xv) \lim_{x \rightarrow 0} \left(\frac{1}{\ln x} - \frac{x}{x - 1} \right) \quad (xvi) \lim_{x \rightarrow 0} \frac{xe^x - \sin x}{3x^2}$$

$$(xvii) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} \quad (xviii) \lim_{x \rightarrow 0} \frac{\sin x - \ln(e^x \cos x)}{x \sin x} \quad (xix) \lim_{x \rightarrow 0} (x)^{\frac{1}{1-x}}$$

i Solⁿ:

$$\lim_{x \rightarrow 0} \frac{x + \sin 2x}{x - \sin 2x}$$

$$= \lim_{x \rightarrow 0} \frac{1 + 2\cos 2x}{1 - 2\cos 2x}$$

$$= \frac{1 + 2}{1 - 2} = -3.$$

ii Solⁿ:

$$\lim_{x \rightarrow 0} \left(\frac{4}{x^2} - \frac{2}{1 - \cos x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{4 - 4\cos x - 2x^2}{x^2(1 - \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{4 - 4\cos x - 2x^2}{x^2 - x^2\cos x}$$

$$= \lim_{x \rightarrow 0} \frac{4\sin x - 4x}{2x - 2x\cos x + x^2\sin x}$$

$$= \lim_{x \rightarrow 0} \frac{4\cos x - 4}{2 + 2x\sin x - 2\cos x + 2x\sin x + x^2\cos x}$$

$$= \lim_{x \rightarrow 0} \frac{-4\sin x}{2x\cos x + 2\sin x + 2\sin x + 2\sin x + 2x\cos x + 2x\cos x - x^2\sin x}$$

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \frac{2x \cos x + 2 \sin x + 2 \sin x + 2 \sin x + 2x \cos x + 2x \cos x - x^2 \sin x}{6 \sin x + 6x \cos x - x^2 \sin x} \\
 &= \lim_{x \rightarrow 0} \frac{-4 \sin x}{6 \sin x + 6x \cos x - x^2 \sin x} \\
 &= \lim_{x \rightarrow 0} \frac{-4 \cos x}{6 \cos x - 6x \sin x + 6 \cos x - x^2 \cos x - 2x \sin x} \\
 &= \lim_{x \rightarrow 0} \frac{-4}{6 + 0 + 6 - 0 - 0} = -\frac{1}{3}
 \end{aligned}$$

$$iii) \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x}}$$

$$= e^{\lim_{x \rightarrow 0} \frac{1}{x} [\frac{\tan x}{x} - 1]}$$

$$= e^{\lim_{x \rightarrow 0} \frac{\tan x - x}{x^2}}$$

$$= e^{\lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{2x}}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{2 \sec^2 x \tan x}{2} \right)}$$

$$= e^{\frac{0}{2}} = 1$$

$$iv) \lim_{x \rightarrow 0} \frac{x - \sin^{-1} x}{\sin^3 x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \frac{1}{\sqrt{1-x^2}}}{3 \sin^2 x \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{2}(-2x)}{(1-x^2)^{3/2} - 3 \sin^3 x}$$

$$= \lim_{x \rightarrow 0} \frac{-x}{(1-x^2)^{3/2} - 3 \sin^3 x}$$

$$= \lim_{x \rightarrow 0} \frac{-(1-x^2)^{3/2} - x \cdot \frac{3}{2}(1-x^2)^{1/2}}{(1-x^2)^3}$$

From MacLaurin's Theorem,

$$\ln(1+x) = x - \frac{x^2}{2!} + \frac{x^3}{3!} - \frac{x^4}{4!} + \dots$$

This can be proved,

$$\lim_{x \rightarrow 0} f(x)^{g(x)} \quad \text{[only if } g(x) \rightarrow \infty \text{]}$$

$$= e^{\lim_{x \rightarrow 0} g(x) [f(x) - 1]}$$

Useful
Shortcut
Formula

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\frac{(1-x^2)^{3/2} - x \cdot \frac{3}{2}(1-x^2)^{1/2}}{(1-x^2)^2}}{6\cos^3 x + 12\sin x \cos x - 9\sin^3 x \cos x} \\
&= \frac{1 - 0}{6 - 0 - 0} = -\frac{1}{6}
\end{aligned}$$

$$\begin{aligned}
v) \quad & \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x} \\
&= \lim_{x \rightarrow \frac{\pi}{2}} e^{[\tan x \ln(\sin x)]} \\
&= \lim_{x \rightarrow \frac{\pi}{2}} e^{\frac{\ln(\sin x)}{\cot x}} \\
&= \lim_{x \rightarrow \frac{\pi}{2}} e^{\frac{\frac{\cos x}{\sin x}}{-\cos^2 x}} \\
&= \lim_{x \rightarrow \frac{\pi}{2}} e^{-\sin x \cos x} \\
&= e^{-0} = 1
\end{aligned}$$

[Here I did not use that shortcut formula. It's preferable not to use that one as sin never taught as that and it's not mentioned in Anton's Book as well. Even if you use that formula try adding a side not like I did in (iii).]

$$\begin{aligned}
vi) \quad & \lim_{x \rightarrow 0} \frac{x - \sin x}{\tan^3 x} \\
&= \lim_{x \rightarrow 0} \frac{1 - \cos x}{3 \tan^2 x \sec^2 x} \\
&= \lim_{x \rightarrow 0} \frac{\sin x}{6 \tan^3 x \sec^2 x + 6 \tan x \sec^4 x} \\
&= \lim_{x \rightarrow 0} \frac{\sin x}{6 \tan x \sec^2 x (\tan^2 x + \sec^2 x)} \\
&= \lim_{x \rightarrow 0} \frac{\sin x \frac{\cos x}{\sin x} \cdot \cos^2 x}{6 (\tan^2 x + \sec^2 x)}
\end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{\cos^3 x}{6(\tan x + \sec^2 x)}$$

$$= \frac{1}{6}$$

vii) $\lim_{x \rightarrow 0} (a^x + x)^{1/x}$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{x} \ln(a^x + x)}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\ln(a^x + x)}{x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\frac{1}{a^x + x} \cdot (a^x \ln a + 1)}{1}}$$

$$= e^{\frac{\ln a + 1}{1}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{a^x \ln a + 1}{a^x + x}}$$

$$= e^{\ln a} \cdot e^1$$

$$= e^{\frac{\ln a + 1}{1}}$$

$$= a \cdot e = ea$$

$$= ea$$

viii) $\lim_{x \rightarrow 0} \frac{\ln \ln(1-x^2)}{\ln \ln \cos x}$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\ln(1-x^2)} \cdot \frac{1}{(1-x^2)} (-2x)}{\frac{1}{\ln(\cos x)} \cdot \frac{1}{\cos x} (-\sin x)}$$

$$= \lim_{x \rightarrow 0} \frac{\ln \cos x}{\ln(1-x^2)} \left(\frac{2x}{\tan x} \right) \left(\frac{1}{1-x^2} \right)$$

$$= \lim_{x \rightarrow 0} \frac{-\tan x}{-\frac{2x}{1-x^2}} \cdot \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{-1} \lim_{x \rightarrow 0} \left(\frac{1}{1-x^2} \right)$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{1-x^2} = \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right) \cdot 2 \lim_{x \rightarrow 0} \left(\frac{x}{1-x^2} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\tan x}{\frac{2x}{1-x^2}} \cdot 1 \cdot 2 \cdot 1$$

$$= \lim_{x \rightarrow 0} \frac{\tan x}{x} \cdot \lim_{x \rightarrow 0} (1-x^2)$$

$$= 1 \times 1 = 1$$

$$ix) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sec^2 x - 2 \tan x}{1 + \cos 4x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \sec^2 x \tan x - 2 \sec^2 x}{-4 \sin 4x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sec^2 x (\tan x - 1)}{-2 \sin 4x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sec x (\sec^2 x) + (\tan x - 1) 2 \sec^2 x \tan x}{-8 \cos 4x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sec^4 x + 2 \sec^2 x \tan^2 x - 2 \sec^2 x \tan x}{-8 \cos 4x}$$

$$= \frac{4 + 2 \times 2 \times 1 - 2 \times 2 \times 1}{-8 \times (-1)}$$

$$= \frac{1}{2}$$

$$x) \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}$$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{x^2} \ln \left(\frac{\sin x}{x} \right)}$$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{x^2} \ln \left(1 + \frac{\sin x - x}{x} \right)}$$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{x^2} \left(\frac{\sin x - x}{x} \right)}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\sin x - x}{x^3}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\cos x}{3x^2}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{-\sin x}{6x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{-\cos x}{6}} = e^{-\frac{1}{6}}$$

Here, I used that formula.
What if I don't want to?

$$\lim_{x \rightarrow 0} e^{\frac{1}{x^2} \ln \left(\frac{\sin x}{x} \right)}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\frac{x}{\sin x} \frac{\cos x - \sin x}{x^2}}{2x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\cos x - \sin x}{2x^2 \sin x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\cos x - x \sin x - \cos x}{4x \sin x + 2x^2 \cos x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{-\sin x - x \cos x}{4x \cos x + 4 \sin x + 4x \cos x - 2x^2 \sin x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{-\cos x + x \sin x - \cos x}{4 \cos x - 4 \sin x + 4 \cos x + 4 \cos x - 4 \sin x - 4 \sin x - 2x \cos x}}$$

$$= e^{\frac{-1 + 0 - 1}{4 - 0 + 4 + 4 - 0 - 0}}$$

$$= e^{\frac{-2}{12}} = e^{-\frac{1}{6}}$$

$$x \rightarrow 0 \quad \sim$$

$$= e^{-\frac{1}{6}}$$

$$xi) \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2}$$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{x^2} \ln \left(1 + \frac{\tan x}{x} - 1 \right)}$$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{x^2} \left(\frac{\tan x - x}{x} \right)}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\tan x - x}{x^3}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\sec^2 x}{3x^2}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{2 \sec^2 x \tan x}{6x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{2 \sec^2 x \tan x + \sec^2 x}{3}}$$

$$= e^{\frac{1}{3}}$$

$$xii) \lim_{x \rightarrow 0} \frac{(e^x - 1) \tan^2 x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{e^x - 1}{x} \times \lim_{x \rightarrow 0} \frac{\tan^2 x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{e^x}{1} \times \lim_{x \rightarrow 0} \frac{2 \tan x \sec^2 x}{2x}$$

$$= 1 \times \lim_{x \rightarrow 0} \frac{\sec^4 x + \tan^2 x \sec^2 x}{1}$$

$$= 1 \times 1 = 1$$

$$= e^{\frac{-2}{12}} = e^{-\frac{1}{6}}$$

This way it's a bit longer
But might be better this way
not use that formula.

$$\lim_{x \rightarrow 0} e^{\frac{\ln(\tan x/x)}{x^2}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{\frac{x}{\tan x} \cdot \frac{x \sec^2 x - \tan x}{x^2}}{2x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{x \sec^2 x - \tan x}{2x^2 \tan x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{x \sec^2 x \tan x + \sec^2 x - \sec^2 x}{4x \tan x + 2x^2 \sec^2 x}}$$

Do it on your own
the rest. The answer
should be same.

$$\lim_{n \rightarrow \infty} \frac{\ln(n)}{n}$$

$$= 1 \times 1 = 1.$$

$$\text{xiii)} \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{x} \ln(1+x)}$$

$$= \lim_{x \rightarrow 0} e^{\frac{2+x}{x}}$$

$$= \lim_{x \rightarrow 0} e^1 = e$$

$$\begin{aligned} \lim_{n \rightarrow 0} e^{\frac{\ln(n)}{n}} \\ = \lim_{n \rightarrow 0} e^{\frac{\frac{1}{1+n}}{1}} \\ = e^1 = e \end{aligned}$$

Even without using that one, this way seemed easier.

$$\text{xiv)} \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin x - x}{x \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - 1}{x \cos x + \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x}{-x \sin x + \cos x + \cos x}$$

$$= \frac{0}{2} = 0$$

$$\text{xv)} \lim_{x \rightarrow 0} \left(\frac{1}{\ln x} - \frac{x}{x-1} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{\ln x} - \lim_{x \rightarrow 0} \frac{x}{x-1}$$

$$= 0 - \frac{0}{0-1}$$

$$= 0$$

$$xvi) \lim_{x \rightarrow 0} \frac{x e^x - \sin x}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{e^x + x e^x - \cos x}{6x}$$

$$= \lim_{x \rightarrow 0} \frac{e^x + e^x + x e^x + \sin x}{6}$$

$$= \frac{1+1+0+0}{6}$$

$$= \frac{1}{3}.$$

$$xvii) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{6x}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x}{6}$$

$$= \frac{1}{6}.$$

$$xviii) \lim_{x \rightarrow 0} \frac{\sin x - \ln(e^x \cos x)}{x \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - \frac{1}{e^x \cos x} (e^x \cos x - e^x \sin x)}{\sin x + x \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - (1 - \tan x)}{\sin x + x \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x + \sec^2 x}{\cos x + \cos x - x \sin x}$$

$$= \frac{-0 + 1}{1 + 1 - 0} = \frac{1}{2}$$

712) $\lim_{x \rightarrow 0} (x)^{\frac{1}{1-x}}$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{1-x} \ln x}$$

$$= \lim_{x \rightarrow 0} e^{\frac{1}{-x}}$$

$$= e^{-\infty} = 0$$